

ResumeIQ – AI-Powered Career Intelligence System

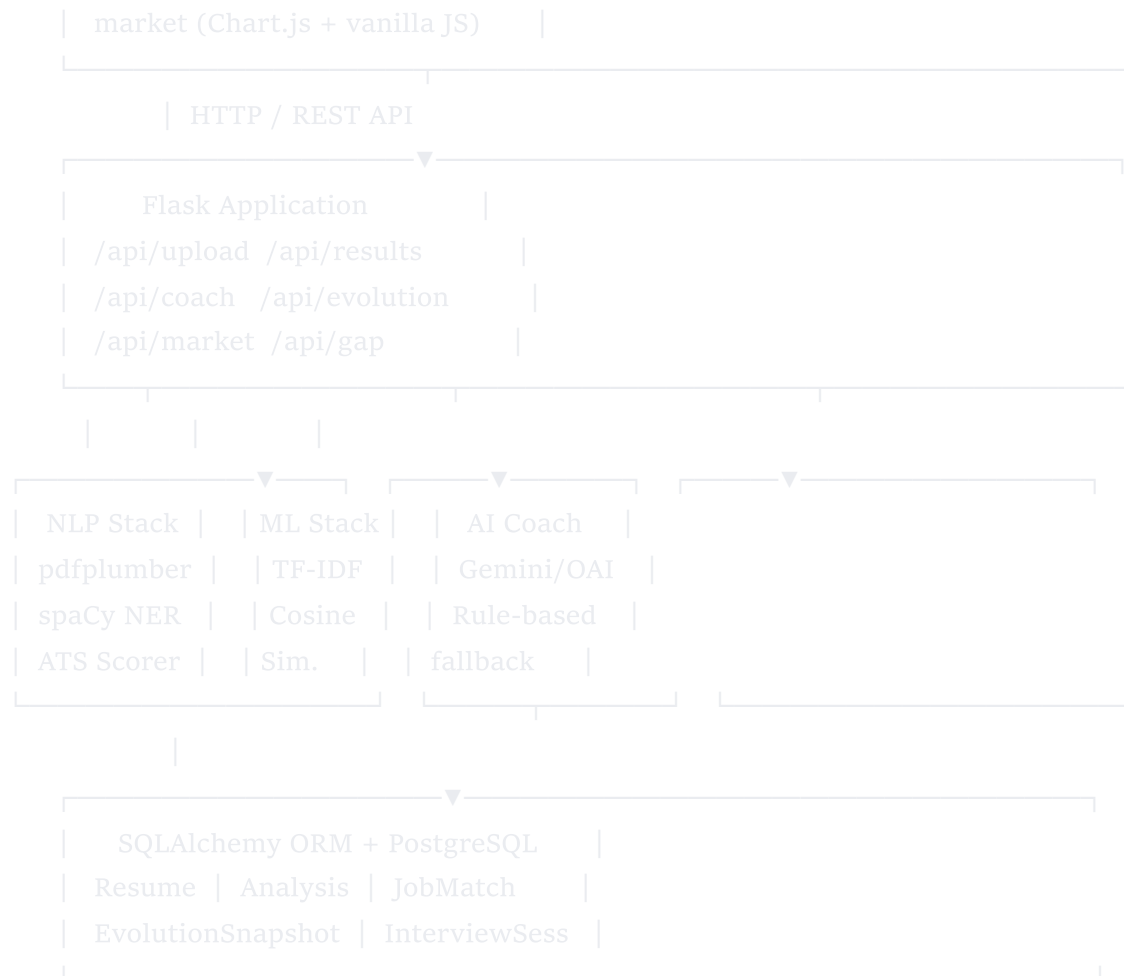
Beyond ATS Scoring. ResumeIQ combines NLP parsing, ML job matching, Career DNA mapping, real-time market pulse, resume evolution tracking, and an AI interview coach – all in one platform.

Novelty Features

Feature	Description	Tech
 Career DNA Map	Maps your skills to 8 career archetypes on a radar chart	Custom TF-IDF + Chart.js
 Resume Evolution Tracker	Tracks score improvement across resume versions	SQLAlchemy + Line Chart
 Real-time Market Pulse	Shows live demand trends vs your current skill set	REST API + Bar/Line Charts
 AI Interview Coach	Generates tailored interview questions from your resume	Gemini/OpenAI API + Fallback

Architecture

User Browser (Frontend) |
index.html → dashboard → coach →



📁 Project Structure

```
resumeiq/
├── backend/
│   ├── app.py          # Flask factory
│   └── api/
│       └── routes.py    # REST API endpoints
```

```
| | └── views.py      # HTML page routes
| | └── parser/
| |   ├── pdf_parser.py  # PDF/DOCX parser + 500-skill taxonomy
| |   ├── ats_scorer.py  # ATS scorer + Career DNA mapper
| |   └── ml/
| |     ├── recommender.py  # TF-IDF cosine similarity recommender
| |     └── ai_coach/
| |       ├── coach.py      # AI interview question generator
| |       └── database/
| |         └── models.py    # SQLAlchemy models
| └── frontend/templates/
|   ├── index.html        # Upload page (dark grid UI)
|   ├── dashboard.html    # Results + DNA + Evolution + Jobs
|   ├── market.html       # Real-time Market Pulse
|   └── coach.html        # AI Interview Coach
| └── docker/
|   ├── Dockerfile
|   └── docker-compose.yml
| └── wsgi.py              # Gunicorn entry
| └── render.yaml          # Render.com deploy config
| └── requirements.txt
| └── .env.example
```

Quick Start (Local)

```
bash
```

```
# 1. Clone and setup
git clone <your-repo>
cd resumeiq
python -m venv venv && source venv/bin/activate

# 2. Install dependencies
pip install -r requirements.txt
python -m spacy download en_core_web_sm

# 3. Configure environment
cp .env.example .env
# Edit .env — add GEMINI_API_KEY if you have one

# 4. Run
python wsgi.py
# Open http://localhost:5000
```

Docker (Local)

```
bash

cd docker
docker-compose up --build
# Open http://localhost:5000
```

Deploy to Render (Free Tier)

1. Push code to GitHub

- 2. Go to render.com → New → Blueprint
- 3. Connect your GitHub repo — Render reads `render.yaml` automatically
- 4. Add env var `GEMINI_API_KEY` (optional) in the Render dashboard
- 5. Click **Deploy** — live in ~5 minutes

 REST API Reference

Method	Endpoint	Description
POST	<code>/api/upload</code>	Upload resume → returns full analysis
GET	<code>/api/results/<id></code>	Get analysis by ID
GET	<code>/api/evolution/<resume_id></code>	Resume version history
POST	<code>/api/coach/<resume_id></code>	Generate interview questions
POST	<code>/api/gap</code>	Skill gap roadmap for target role
GET	<code>/api/market</code>	Market pulse trend data
GET	<code>/api/health</code>	Health check

Example: Upload Resume

```
bash
```

```
curl -X POST http://localhost:5000/api/upload \  
-F "resume=@my_resume.pdf" \  
-F "note=Added AWS certification"
```

Example: Generate Interview Questions

```
bash  
  
curl -X POST http://localhost:5000/api/coach/1 \  
-H "Content-Type: application/json" \  
-d '{"target_role": "Data Scientist", "num_questions": 10}'
```

Tech Stack

Layer	Technologies
Backend	Python, Flask, SQLAlchemy, Gunicorn
NLP/Parsing	spaCy, pdfplumber, python-docx, regex
ML	TF-IDF, Cosine Similarity, scikit-learn
GenAI	Google Gemini 1.5 Flash / OpenAI GPT-4o-mini
Database	SQLite (dev), PostgreSQL (prod)
Frontend	HTML5, CSS3, Vanilla JS, Chart.js 4
DevOps	Docker, Docker Compose, Render

Career DNA Archetypes

ResumeIQ maps your skills to 8 archetypes and renders a radar chart:

-  **The Builder** — Backend & API specialist
-  **The Data Wizard** — ML & statistical modelling
-  **The Cloud Architect** — AWS/GCP/Azure & IaC
-  **The Data Engineer** — Pipelines & warehousing
-  **The Storyteller** — Visualisation & BI
-  **The AI Whisperer** — LLMs, RAG & GenAI
-  **The Full-Stack Engineer** — React, Node, APIs
-  **The DevOps Ninja** — CI/CD, Kubernetes, SRE

ATS Scoring Weights

Component	Weight	Description
Skill Coverage	30%	Skills found vs 500-skill taxonomy
Section Completeness	25%	Summary, Experience, Skills, Education
Experience	20%	Years of experience (log curve)
Keyword Density	15%	Action verbs and industry terms
Contact Info	10%	Email, phone, LinkedIn, GitHub

